

Comparison of measured and computed neutron yield from PF1000 plasma focus device operated with deuterium gas

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ABSTRACT

The measured current wave shapes of PF1000 plasma focus discharged at 16 kV in deuterium in the pressure range 1.05–1.2 Torr are fitted to the computed current wave shapes using the Lee model code. Very good fit is achieved which suggests that the code is configured very well for the PF1000 for these shots. This is borne out by the computed neutron yield arising from each shot agreeing well with the measured yield. The computed average yield of 8 shots at 1.2 Torr is 1.78×10^9 neutrons per shot (n/sh) compared to the measured average yield of $(1.75 \pm 0.2) \times 10^9$ (n/sh). The computed average yield of 16 shots at 1.05 Torr is 2.33×10^9 (n/sh) compared also to the measured average of $(2.29 \pm 0.2) \times 10^9$ (n/sh). The conclusion is that the neutron production of the PF1000 is well estimated by the Lee Code simply by fitting the computed current wave shape to the measured current wave shape.

1. Introduction

Plasma focus devices are effective sources of high-energy electrons and ions and in the event of the presences of deuterium in the load of the neutrons from D-D fusion reaction are produced (Bernard et al., 1998; Krauz, 2006; Gribkov et al., 2007a,b; Sadowski and Scholz, 2008; Krishnan, 2012). Deuterium plasma has been extensively studied using the PF-1000 plasma focus device with the multi-frame laser interferometry, X-ray, and neutron diagnostics, and the correlation between neutron production and transformation of the plasmoids and constrictions has been proven (Kubes et al., 2011). Neutron yields in order of 10^{10} /discharge have been obtained in PF1000 plasma focus device (1 MA of the peak current) with the deuterium filling. It was found that ordered structure are generated in the pinch column which could be due to closed internal currents interacting with external currents and the resultant magnetic fields. The currents could take the form of filaments and magnetic reconnections are speculated upon as the possible cause of particle acceleration (Kubes et al., 2019a,b). Furthermore, the neutron yield versus the current and stored energy for different plasma focus devices has been studied and scaling rules have also been deduced (Lee,

2008; Lee, 2009). The neutron production from various plasma focus machines change according to electrode geometry (Moreno et al., 2000), gas pressure (Marciniak et al., 2018) and stored energy (Soto, 2005). Nevertheless, the neutron yield optimization has also been investigated by finding the optimum electrode shape and geometry (Bures et al., 2010; Verma et al., 2009; Babazadeh et al., 2002) and D₂ gas pressures (Saw et al., 2015). Based on the published current waveform and information on neutron yield at a typical pressure of 3.5 Torr (Gribkov et al., 2007a,b), the Lee model code has been utilized to compute the neutron yield of PF1000 as a function of pressure (Saw et al., 2014). In addition to that, the computational study on D-D and D-T (1:1) fusion for PF1000 plasma focus parameters is also achieved using Lee model (Rakib et al., 2020). The neutron yield performance of the D-T gas mixture is compared to that of the D gas and the optimum gas pressure for the highest neutron yield is identified. As mentioned above, many discussions of the PF1000 neutron yields have previously been confined to numerical Experiments linked to the actual experiments by a measured current trace and compared to a range of neutron yields. There have not been shot by shot comparisons. The aim of our present paper is to compare the neutron yield of actual PF1000 shots to the

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numerical equivalent shots estimated by the Lee model code for different experimental conditions.

2. Experimental device and diagnostics

The PF-1000 plasma focus has 480 mm long coaxial electrodes. The copper anode comprises a tube of diameter 231 mm. The cathode was formed of twelve 8-cm diameter stainless-steel tubes positioned symmetrically around the axis of the coaxial electrodes. For these Experiments the capacitor bank ($C_0 = 1332 \mu\text{F}$) was operated at 16 kV at 170.5 kJ. Maximal discharge current amounted to (1100–1300) kA. The operational pressures were 1.05 and 1.2 Torr in deuterium. The voltage and current-derivative waveforms, as well as the total current intensity traces, were measured at the main collector plate. Three scintillation detectors were used to record signals at a distance of 7 m from the focus center in the downstream (at 0°), upstream (at 180°), and side-on (at 90°) directions. These measurements allowed us to determine the instants of the emission of hard X-rays (HXR) and fusion-produced neutrons. The mean values of the energy of the fusion neutrons and primary deuterons were determined with the time-of-flight method from the temporal difference in the amplitudes of the neutron signals recorded downstream and upstream. The total neutron yield was measured using two calibrated silver activation counters, which were placed outside the main experimental chamber. These counters were calibrated using an Am–Be source placed on the anode axis. The uncertainty in the neutron yield measurements were estimated as ± 0.2 . In all cases, the maximum of the dip in the current derivative was assigned as the instant $t = 0$. This corresponded to the formation of the first plasmoid and a peak of the soft x-ray (SXR) signal. The time from the breakdown to the derivative dip was about 7 μs . The evolution of the constriction started at 50–100 ns. The formation of secondary plasmoids and the decay of the constriction were usually recorded 100 ns–200 ns after the dip. The uncertainty in the timing of different measuring channels during a single discharge was about 3 ns – 5 ns. The results following from these campaigns were published in (Kubes et al., 2020a,b). The dominant HXR and neutron production usually starts during the decay of the plasmoids and the disruption of the constriction. The narrow constriction often decays axially when it is absorbed into the secondary plasmoids or by the formation of new third-generation plasmoids. The zippering of the column along its length leads to the temporal evolution of the final phase, in which there are a few constrictions on top of each other. The plasmoids decay when their centers explode and their boundaries expand. Then, some of the magnetic energy conserved in the closed internal and external structures can accelerate fast particles. The release of the magnetic energy can also proceed by magnetic reconnections during the spontaneous coalescence of two parallel filaments, in which the decaying magnetic field induces a strong electric field. Then, the magnetic energy also increases and its total value in the volume above the anode front can reach several kilojoules (for several hundreds of nanoseconds). This energy can be contained in the closed local magnetic fields and it can be released as energetic particle beams, which are emitted from the plasma column (mainly during its decay), as well as from its surroundings. This energy of about several kilojoules was estimated from the measured neutron yield, the mean energy of the accelerated fast deuterons, the cross section of D–D fusion reactions for this deuterium energy, and the known density of the dense plasma at the top of the dense pinch column. The mean energy of fast deuterons was estimated from the temporal differences in the neutron signals recorded in the downstream and upstream directions. The mean energy can reach 5 kJ. The detailed analyzes, studies, and discussions of hard x-ray, neutron production, plasmoid formation, currents, magnetic fields, scenario of a magnetic dynamo and magnetic reconnection in the PF1000 plasma focus discharges were presented in earlier papers (Kubes et al., 2019, 2020, 2020; Kubes et al., 2020a,b).

2.1. The Lee model code for the computation

The 5-phase Lee code is a relatively simple, semi-empirical, efficient, hybrid computer code simulating approximately all main phases of discharge in the plasma focus. It is energy, mass, momentum and charge consistent. It uses a realistic power source model (discharge in a dynamic and damped equivalent RLC circuit). It assumes the validity of Bennett relation in each small time step of the code in the equilibrium pinch phase of discharge formation of plasma pinch in form of a column in the dynamic pressure equilibrium and thermodynamic equilibrium with uniform/average plasma/discharge parameters along dimensions. It includes the finite velocity of small disturbances propagation in plasma during radial collapse and compression taken as the acoustic/sound velocity (to account for the contact between the current sheath and the shock wave during radial adiabatic implosion before stable pinch formation and to approximately calculate the lifetime of plasma pinch in the equilibrium state). The code includes also the ionization effects in plasma, the shock-wave heating model in radial phases of discharge before stable pinch formation, the classical model of plasma column ohmic heating based on the Spitzer resistivity, the anomalous resistivity and heating due to the micro-instabilities development. The code enables also x-ray bremsstrahlung, line and recombination radiation emission from an equilibrium plasma column and couples it with the plasma pinch compression dynamics. The plasma self-absorption and transition from volumetric to surface emission is also included as well as the neutron emission based on both thermonuclear and beam-target fusion models. In the Lee model, a beam of fast deuteron ions is produced by a vacuum diode in a thin layer close to the anode, with plasma disruptions generating the necessary high voltages. The inductively generated deuteron beam traverses the pinch column of deuterium plasma to produce the fusion neutrons. The beam-target production (Y_{b-t}) was deduced (Lee., 2014, 2020; Lee and Saw, 2017) as:

$$Y_n = Y_{b-t} = C_n n_i I_{\text{pinch}}^2 z_p^2 (\ln(b/r_p)) \sigma U^{1/2} \quad (1)$$

where, n_i is the maximum pinch ion density per discharge (average and uniform along pinch dimensions); I_{pinch} - the initial current through the pinch (uniform along pinch dimensions); r_p and z_p - the final pinch radius and length; b - the cathode radius; σ - the cross-section of D–D fusion reaction (for the beam energy equal to three times maximum voltage induced- V_{max}), n -branch; and U - the disruption-caused diode voltage. Data fitting gave $U = 3 V_{\text{max}}$, where V_{max} is the maximum voltage induced by the radially collapsing current sheet; and $C_n = 8.54 \times 10^8$ (all quantities in SI units) is a constant which was calibrated at an experimental point of 0.5 MA from a graphical presentation of all available measured Y_n data from different plasma focus devices (global constant).

The Lee model code simulates the total current. Four model parameters are used representing the mass swept-up factor (f_m) and the plasma current factor (f_c) for the axial phase as well as the mass swept-up factor (f_{mr}) and the plasma current factor (f_{cr}) for the radial phases. These four model parameters are used to account for the experimentally observed mass loss and current loss in an axial and the radial phase of discharge, respectively (percentages of total mass and current flowing between electrodes in axial and radial phases). After each run, the user changes one of the model parameters so as to improve the fitting of the computed to the measured current trace for the next run. This fitting procedure is done sequentially starting with f_m and f_c whilst inspecting the fit of the current trace in the axial phase before the current dip, and then proceeds to f_{mr} and f_{cr} whilst inspecting the fit of the current dip. Additionally, static inductance L_0 and stray resistance r_0 parameters may be fine-tuned from run to run during the fitting procedure of current traces. In general, it is well known that the current signal from a plasma focus is one of the best performance indicators that can be obtained. Some of the most important information that can be derived using Lee model, like the axial and radial phase dynamics as well as radiation yields, are dependent on the current flow in the plasma. Moreover, once fitted, the code

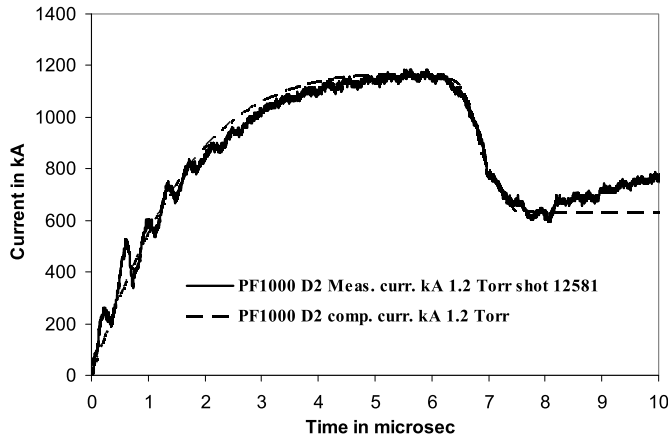


Fig. 1. Computed and measured currents of the PF1000 at 16 kV, 1.2 Torr of deuterium (shot 12581).

outputs the following realistic data: dynamics and energy content of the phases, pinch geometry, temperatures and densities, line radiation and neutron yields. In this article, all experimental discharges (experimental current traces) were fitted individually with simulated discharges (simulated current traces) using four model parameters as well as L_0 and r_0 parameters. The code has a control panel into which are input the PF1000 bank parameters, the tube parameters and the operation parameters, thus configuring the code to work as PF1000.

3. Experiments and simulations on PF1000 plasma focus device: Results and discussion

Several Experiments are performed on the ($E_0 \sim 170.5$ kJ) PF1000 in deuterium (session of experiments in September–October 2018) at pressures of 1.2 Torr (shots: 12581–12589) and 1.05 Torr (shots: 12590–12606). In this work, we use the measured current wave shapes to calculate the neutron yields using the Lee model at the same experimental conditions. The fitting procedures using Lee model (Lee, 2014) are investigated for each measured current step by step. For this purpose, the Lee model code is configured as the PF1000 using the following parameters as an example, for shot 12581:

Bank: $L_0 = 25$ nH, $C_0 = 1332$ μ F, $r_0 = 6.1$ m Ω ;

Tube: $b = 16$ cm, $a = 11.55$ cm, $z_0 = 48$ cm;

Operational: $V_0 = 16$ kV, $p_0 = 1.2$ Torr, deuterium gas

where L_0 , C_0 , b , a , z_0 , V_0 , p_0 are respectively the static inductance (nominal), the storage capacitance, the cathode radius, the anode radius, the anode length, the operating voltage and initial pressure.

The measured wave shape of the current is then used as the basis to fit the computed current. The four model parameters derived from the fitting are the axial phase mass swept-up factor $f_m = 0.17$, and corresponding plasma current factor $f_c = 0.7$ and the radial phase mass swept-up factor $f_{mr} = 0.25$ and corresponding current factor $f_{cr} = 0.75$. Fig. 1 shows that the fitting is very good until the end of the current dip (at around 8 μ s), beyond which no fitting was necessary, hence no fitting was attempted.

With these fitted model parameters, the code outputs the following results: the peak axial speed is $V_a = 10.5$ cm/ μ s, the shock speed (on-axis) is $V_s = 22$ cm/ μ s; the peak radial piston speed is $V_p = 18$ cm/ μ s; the peak value of total discharge current I_{peak} is about 1165 kA; the pinch current I_{pinch} is 523 kA, the speed factor ($SF = (I_{peak}/a)/p_0^{1/2}$) is 92 kA/cm per [Torr of D_2] $^{1/2}$, the final pinch radius reaches 2.3 cm, the pinch length is 18.2 cm, the maximum voltage induced by the radially collapsing current sheet V_{max} is 30 kV and the pinch duration is about 212 ns. At these conditions, the neutron yield (Y_n) is computed, to be

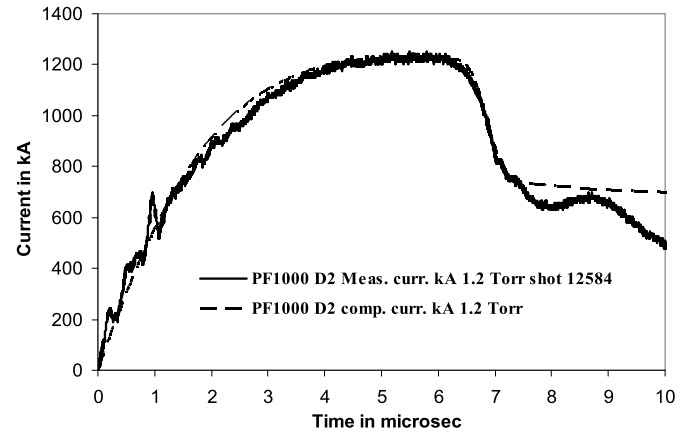


Fig. 2. Computed and measured currents of the PF1000 at 16 kV, 1.2 Torr of deuterium (shot 12584).

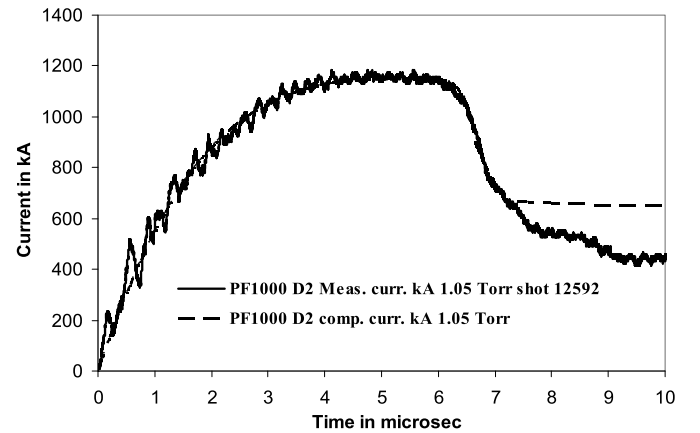


Fig. 3. Computed and measured currents of the PF1000 at 16 kV, 1.05 Torr of deuterium (shot 12592).

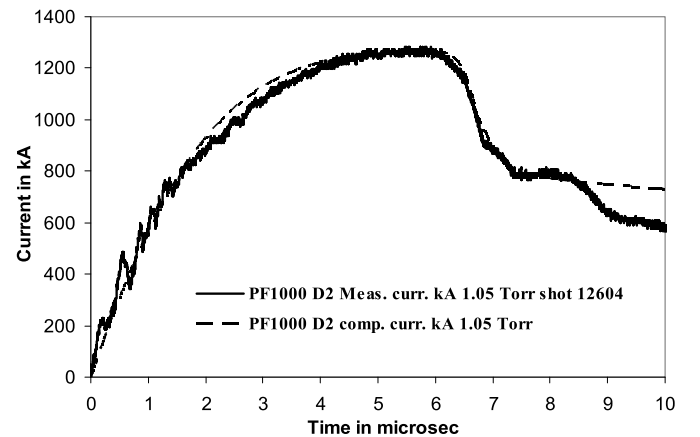


Fig. 4. Computed and measured currents of the PF1000 at 16 kV, 1.05 Torr of deuterium (shot 12604).

6.14×10^9 n/discharge, which is well compared with the total measured neutron yield of $(6.1 \pm 0.2) \times 10^9$ (n/sh).

In the same manner, we fitted the other measured current traces of D_2 PF1000 discharges at 1.05 and 1.2 Torr (see Figs. 2–4) and the obtained fitting parameters are presented in the Table 1.

Table 1 presents that the fitted static inductance L_0 has a value of 25 nH for all fitted discharges, while the resistance r_0 ranges between 4 and

Table 1

The PF1000 plasma focus discharges under consideration are displayed in the first column. The remaining columns are the Deuterium pressure p_0 , the static inductance L_0 , short-circuited resistance r_0 , peak current I_{peak} , pinch current I_{pinch} and the four fitted model parameters.

PF1000 Shots	p_0 (Torr)	L_0 (nH)	r_0 (m Ω)	I_{peak} (kA)	I_{pinch} (kA)	f_m	f_c	f_{mr}	f_{cr}
12581	1.20	25.0	6.1	1165	523	0.17	0.70	0.26	0.75
12582	1.20	25.0	4.2	1303	389	0.19	0.70	0.10	0.43
12583	1.20	25.0	4.0	1335	390	0.21	0.70	0.10	0.42
12584	1.20	25.0	5.3	1228	404	0.19	0.70	0.10	0.50
12586	1.20	25.0	5.1	1237	365	0.19	0.70	0.08	0.43
12587	1.20	25.0	6.5	1135	413	0.17	0.70	0.13	0.57
12588	1.20	25.0	5.0	1285	312	0.24	0.70	0.03	0.35
12589	1.20	25.0	4.3	1321	312	0.22	0.70	0.03	0.33
12590	1.05	25.0	4.4	1314	368	0.24	0.70	0.10	0.40
12592	1.05	25.0	6.1	1151	438	0.18	0.70	0.17	0.60
12593	1.05	25.0	5.9	1192	262	0.22	0.70	0.03	0.30
12594	1.05	25.0	5.4	1224	332	0.22	0.70	0.04	0.40
12595	1.05	25.0	4.6	1269	391	0.21	0.70	0.12	0.45
12596	1.05	25.0	6.5	1131	434	0.18	0.70	0.12	0.62
12597	1.05	25.0	5.1	1231	449	0.20	0.70	0.15	0.57
12598	1.05	25.0	6.0	1169	302	0.20	0.70	0.04	0.37
12599	1.05	25.0	5.3	1223	304	0.21	0.70	0.04	0.35
12600	1.05	25.0	4.1	1328	296	0.24	0.70	0.04	0.30
12601	1.05	25.0	4.8	1263	348	0.22	0.70	0.06	0.40
12602	1.05	25.0	5.2	1214	422	0.19	0.70	0.14	0.53
12603	1.05	25.0	5.3	1207	598	0.19	0.70	0.35	0.85
12604	1.05	25.0	4.6	1261	415	0.20	0.70	0.18	0.48
12605	1.05	25.0	4.7	1241	489	0.19	0.70	0.27	0.61
12606	1.05	25.0	4.5	1268	457	0.20	0.70	0.24	0.54

Table 2

The computed and measured neutron yields of PF1000 plasma focus device at D_2 1.05 and 1.2 Torr. The peak axial speed (V_a), the shock speed (V_s), the peak radial piston speed (V_p), the maximum pinch density (n_i), r_p and z_p -the final pinch radius and length.

PF1000 Shots	p_0 (Torr)	V_a (cm/ μ s)	V_s (cm/ μ s)	V_p (cm/ μ s)	n_i (10^{23} m $^{-3}$)	r_p (cm)	z_p (cm)	Computed neutron yield (n/sh)<	Measured neutron yield (n/sh)
12581	1.20	10.5	22.2	18.0	1.7	2.40	18.2	6.14×10^9	$(6.1 \pm 0.2) \times 10^9$
12582	1.20	11.1	24.6	18.6	0.8	2.17	16.6	1.51×10^9	$(1.5 \pm 0.2) \times 10^9$
12583	1.20	10.8	24.6	18.7	0.8	2.17	16.6	1.54×10^9	$(1.5 \pm 0.2) \times 10^9$
12584	1.20	10.5	26.1	20.4	0.7	2.24	16.6	1.55×10^9	$(1.5 \pm 0.2) \times 10^9$
12586	1.20	10.7	25.9	19.8	0.6	2.18	16.3	1×10^9	$(1 \pm 0.2) \times 10^9$
12587	1.20	10.4	24.0	19.0	0.9	2.28	17.0	1.83×10^9	$(1.8 \pm 0.2) \times 10^9$
12588	1.20	9.8	35.7	27.6	0.2	2.13	15.4	3.2×10^8	$(3 \pm 0.2) \times 10^8$
12589	1.20	10.5	35.3	26.9	0.3	2.10	15.3	3.3×10^8	$(3 \pm 0.2) \times 10^8$
12590	1.05	10.4	24.1	18.2	0.7	2.15	16.5	12×10^8	$(10 \pm 0.2) \times 10^8$
12592	1.05	10.8	23.9	18.9	1.0	2.30	17.4	2.53×10^9	$(2.5 \pm 0.2) \times 10^9$
12593	1.05	10.2	31.2	23.3	0.2	2.05	15.3	1.7×10^8	$(1 \pm 0.2) \times 10^8$
12594	1.05	10.4	35.7	27.9	0.3	2.18	15.6	4.2×10^8	$(4 \pm 0.2) \times 10^8$
12595	1.05	11.0	24.8	18.9	0.8	2.19	16.8	1.53×10^9	$(1.5 \pm 0.2) \times 10^9$
12596	1.05	10.6	28.2	22.8	0.7	2.33	16.9	1.85×10^9	$(1.8 \pm 0.2) \times 10^9$
12597	1.05	10.8	25.8	20.4	0.9	2.30	17.2	2.64×10^9	$(2.6 \pm 0.2) \times 10^9$
12598	1.05	10.6	32.0	24.5	0.3	2.13	15.6	3.1×10^8	$(3 \pm 0.2) \times 10^8$
12599	1.05	10.6	32.0	24.3	0.3	2.11	15.6	3.1×10^8	$(3 \pm 0.2) \times 10^8$
12600	1.05	10.8	30.5	22.6	0.3	2.05	15.5	3.3×10^8	$(3 \pm 0.2) \times 10^8$
12601	1.05	10.7	31.7	24.5	0.4	2.17	15.9	6.2×10^8	$(6 \pm 0.2) \times 10^8$
12602	1.05	11.1	24.8	19.3	0.9	2.26	17.1	2.11×10^9	$(2 \pm 0.2) \times 10^9$
12603	1.05	11.0	23.6	19.5	1.9	2.46	18.8	11.1×10^9	$(11.2 \pm 0.2) \times 10^9$
12604	1.05	11.2	21.1	15.9	1.2	2.20	17.5	2.54×10^9	$(2.5 \pm 0.2) \times 10^9$
12605	1.05	11.4	20.9	16.3	1.7	2.30	18.3	5.52×10^9	$(5.5 \pm 0.2) \times 10^9$
12606	1.05	11.2	20.4	15.5	1.6	2.25	18.0	4.20×10^9	$(4.1 \pm 0.2) \times 10^9$

6.5 m Ω . These results reflect the stable regime of the PF1000 operation under these experimental conditions. It is also noticed that the f_m , f_c , f_{mr} and f_{cr} model parameters (required to obtain the good fit) shift from shot to shot with an average values as follow: 0.2, 0.7, 0.12 and 0.48, respectively. This is attributed to the processes variations occurring within each PF1000 discharge, including the dynamics, energetics and thermodynamics. The above mentioned fitted parameters together with the bank, tube and operating parameters for each shot give us the plasma focus parameters (velocities & densities) as well as the neutron yields at 1.05 and 1.2 Torr of D_2 (Table 2). The good agreement between the computed and measured neutron yields is evident from the Table 2 and Figs. 5 and 6.

The very good fit that is achieved for the computed current waveform and the measured current waveform (Figs. 1–4) suggests that the code is configured very well for the PF1000 for these shots. It is worth mentioning that neutron yields in plasma focus machines are known to have unavoidable shot-to-shot variation even when all parameters are pre-set to be as accurately reproducible as possible. The most important contribution to such variation is likely the random changes in conditions of the electrodes from shot-to-shot. Indeed the data of Tables 1 and 2 show better reproducibility than most other machines in both dynamics (the speeds) and neutron yield. In previously studies, as is also observed in our present data of Table 1, we had indications that over such a small range of pressures i.e. from 1.05 to 1.2 Torr there is no noticeable

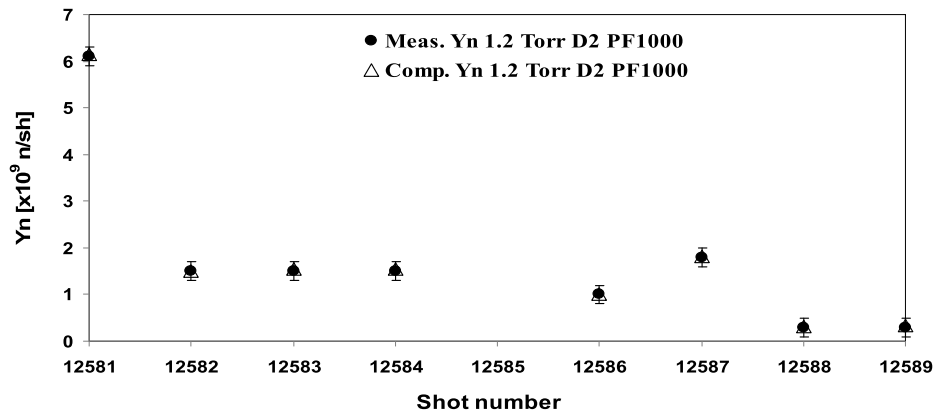


Fig. 5. Calculated and measured neutron yields of the PF1000 at 16 kV, 1.2 Torr of deuterium.

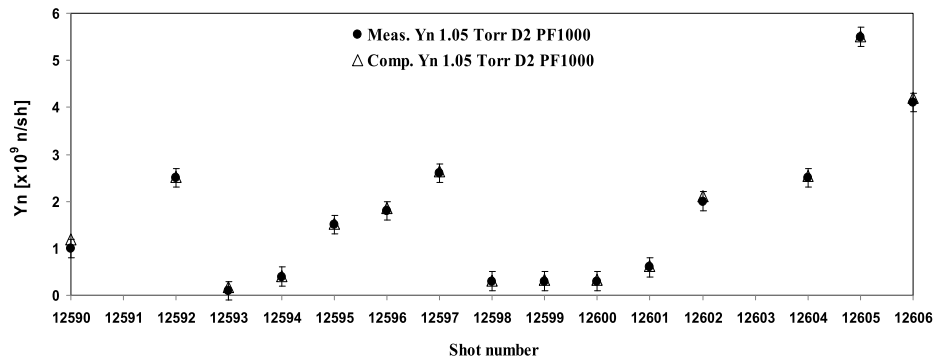


Fig. 6. Calculated and measured neutron yields of the PF1000 at 16 kV, 1.05 Torr of deuterium.

correlation in the 4 model parameters with the operating pressure. Within a series of shots at one pressure the 3 model parameters f_m , f_{mr} and f_{cr} also exhibit shot-to-shot variations. The model parameter f_c is kept constant at 0.7 in order to simplify the fitting process.

This is borne out by the computed neutron yield arising from each shot agreeing well with the measured yield (Figs. 5 and 6). The computed average yield of 8 shots at 1.2 Torr is 1.78×10^9 (n/sh) compared to the measured average yield of $(1.75 \pm 0.2) \times 10^9$ (n/sh) differing by less than 2% although one of the lowest yield shots showed a difference of 10% between the computed and measured yields. The measured yield ranges from $(3 \pm 0.2) \times 10^8$ to $(6.1 \pm 0.2) \times 10^9$ (n/sh). The computed average yield of 16 shots at 1.05 Torr is 2.33×10^9 (n/sh) compared to the measured average of $(2.29 \pm 0.2) \times 10^9$ (n/sh) differing also by less than 2% although the lowest yield shot showed a difference of 70%. The measured yield ranges from $(1.7 \pm 0.2) \times 10^8$ to $(11.1 \pm 0.2) \times 10^9$ (n/sh).

4. Conclusion

The measured current waveforms of PF1000 plasma focus operating at 16 kV in deuterium (1.05–1.2 Torr) are fitted to the computed currents using the Lee model code. Very good fit is achieved which suggests that the code is configured very well for the PF1000 for these shots. This is borne out by the computed neutron yield arising from each shot agreeing well with the measured yield. The computed average yield of 8 shots at 1.2 Torr is 1.78×10^9 (n/sh) compared to the measured average yield of $(1.75 \pm 0.2) \times 10^9$ (n/sh) and the computed average yield of 16 shots at 1.05 Torr is 2.33×10^9 (n/sh) compared also to the measured average of $(2.29 \pm 0.2) \times 10^9$ (n/sh). The conclusion is that the neutron yield of the PF1000 is well estimated by the Lee Code simply by fitting the computed current waveform to the measured current waveform.

Author statement

Mohamad Akel: Conceptualization, Methodology, Software, Writing-Original draft preparation, Pavel Kubes: Validation, Investigation, Writing- Reviewing and Editing, Marian Paduch: Investigation, Sing Lee: Software, Writing- Reviewing and Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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